

Four cell-filter models

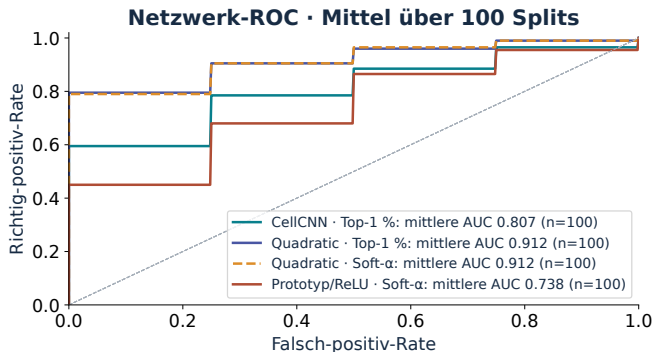
Same data, same donor splits, same half-max rule

Modell	Zellresponse $h_k(x)$	Pooling
CellCNN	$\text{ReLU}(b_k + w_k^\top x)$	hartes Top-1 %
Quadratic Top-1 %	$\text{ReLU}(b_k + w_k^\top x + \sum_j q_{kj} x_j^2)$	hartes Top-1 %
Quadratic Soft- α	$\text{ReLU}(b_k + w_k^\top x + \sum_j q_{kj} x_j^2)$	learnable Soft- α
Prototyp Soft- α	$\text{ReLU}(\rho_k - \frac{1}{d} \sum_j a_{kj} (x_j - c_{kj})^2)$	learnable Soft- α

All models aggregate cell responses, then apply a linear classifier.

Network classification and disease-association of the selected population are scored separately.

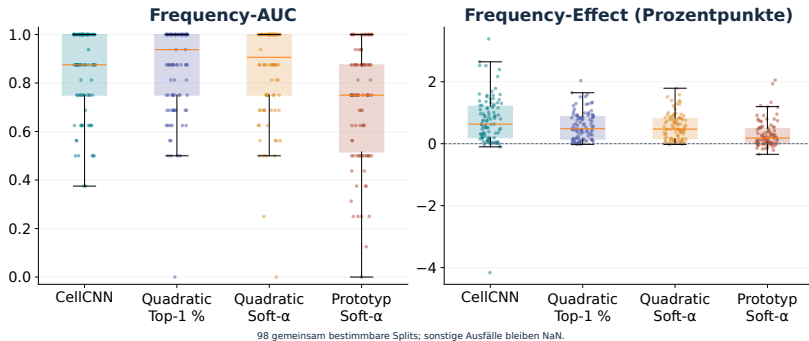
Network classification: quadratic filters win



Model	Mean	Median
CellCNN	0.807	0.875
Quad. Top-1%	0.913	1.000
Quad. Soft- α	0.913	1.000
Prototype	0.738	0.750

Highest mean network AUC: Quadratic Top-1% and Quadratic Soft- α (0.913).

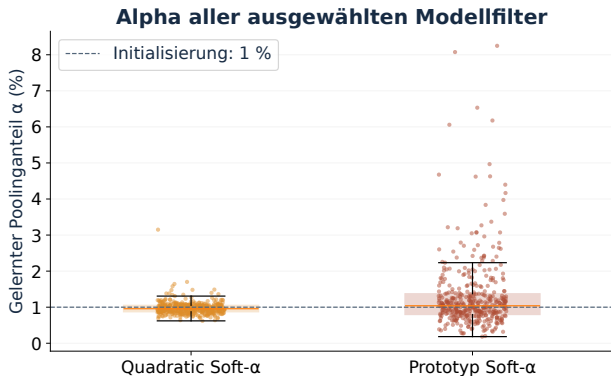
Frequency AUC and effect size answer different questions



Largest mean frequency effect: CellCNN (0.740 percentage points).

AUC: donor ranking. Effect: difference of mean population frequencies; unit percentage points.

What does the extra pooling parameter learn?



Rank fraction, not population frequency

Median α : Quadratic 0.96%; Prototyp 1.04%. Alpha describes pooling, not the measured cell frequency.

Take-away

- Quadratic geometry (hard or soft) gives the best network classification here.
- Learnable soft pooling could add further gain for different data
- Quadratic Soft- α beats CellCNN in 53/100 paired splits (mean $\Delta\text{AUC} +0.105$).
- Limitation: only 20 independent donors – more splits are not more independent evidence.

Quadratic Hard and Soft both reach 0.913 mean network AUC; learnable soft pooling doesn't increase it here.

Backup: selecting a filter and defining half-max population frequency

Filter selection and fixed training threshold

$$\Delta_k = V_{1k} - V_{0k}, \quad k^* = \arg \max_{k: \Delta_k > 0} \Delta_k$$

$$M = \max_{x \in \text{inner-train cells}} h_{k^*}(x), \quad t = \frac{1}{2} M$$

$$f_i = \frac{\#\{x \in \text{test sample } i : h_{k^*}(x) > t\}}{N_i}$$

Same fixed sample of up to 20,000 cells per test donor.
The strict rule also applies when $M = 0$.

Two population metrics

$$\text{AUC}_{\text{freq}} = \text{AUC}(y_i, f_i)$$

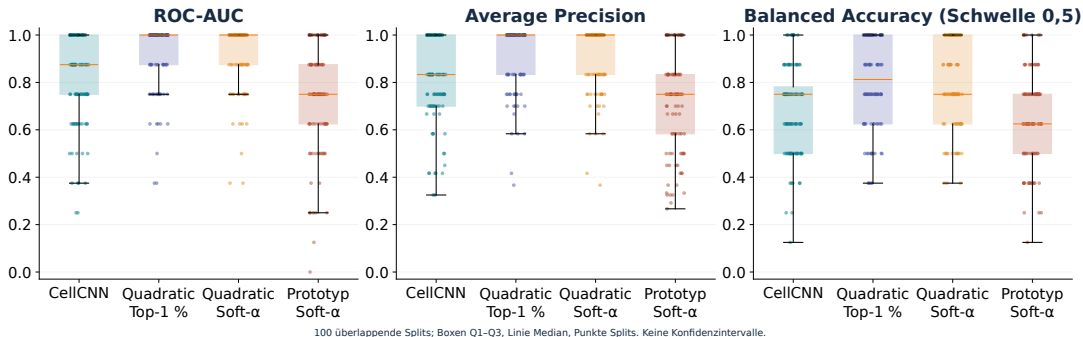
$$\text{Effect} = \bar{f}_{\text{CMV}+} - \bar{f}_{\text{CMV}-}$$

Without a positive filter, values remain missing. No post-hoc flipping of AUC direction.

For frequency comparisons: **98** jointly determinable splits.

Half-max is a cell-selection rule, not clustering; CMV is a donor label, not a cell-type label.

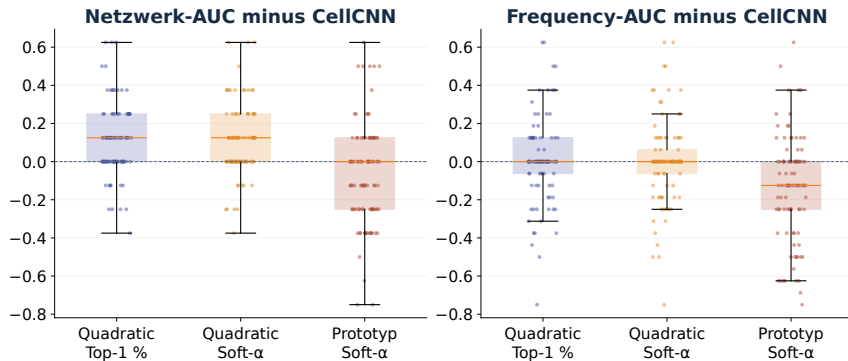
Backup: how much does classification vary between splits?



Highest means: AP for Quadratic Top-1% (0.907); BA for Quadratic Top-1% (0.801).

ROC-AUC, Average Precision and Balanced Accuracy per split; BA at score threshold 0.5. Boxes Q1-Q3, line = median; spread is not a confidence interval.

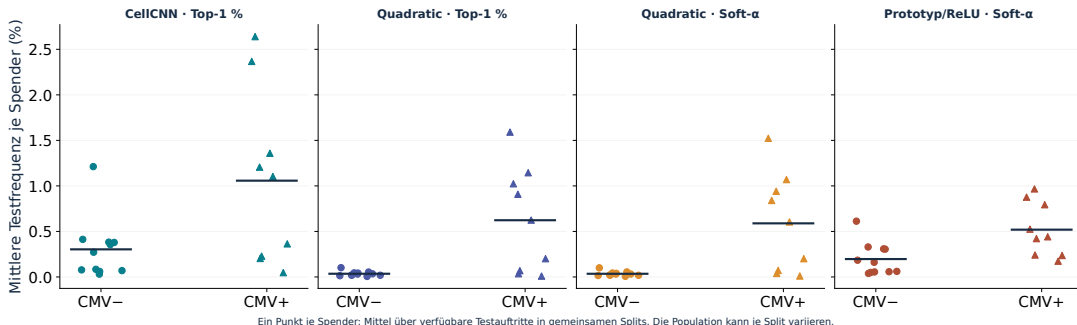
Backup: paired differences to the linear CellCNN baseline



Quadratic Soft- α vs. CellCNN: 53 better, 37 tied, 10 worse; mean difference +0.105.

Differences computed within the same outer split. Positive values mean a higher AUC than CellCNN.

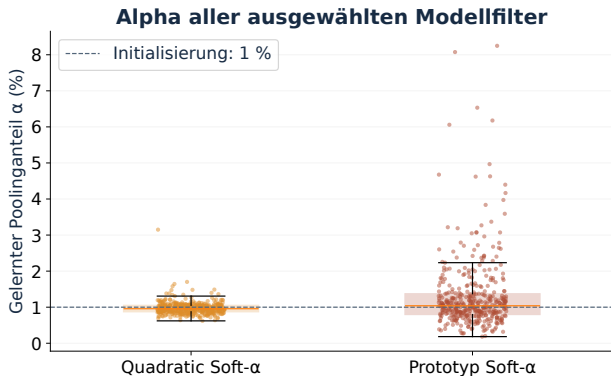
Backup: what frequencies do individual donors show?



Each donor counts as one point; repeated test runs are averaged per donor first.

Circles: CMV-; triangles: CMV+. Bars: mean of donor points. A different filter may define the population in each split.

Backup: what does the extra pooling parameter learn?



Rank fraction, not population frequency

$$\alpha_k = \sigma(\theta_k), \quad \alpha_k^{(0)} = 0.01$$

$$m_{rk} = \sigma\left(\frac{\alpha_k - q_r}{0.002}\right)$$

With $q_r = (r - \frac{1}{2})/N$. Temperature stays fixed. A soft α of 1% is not identical to hard top-1%.

Median α : Quadratic 0.96%; Prototyp 1.04%. Alpha describes pooling, not the measured cell frequency.

Backup: scope of the controlled comparison

- **Quadratic Hard vs. Soft:** same cell response and L2 penalty; only the pooling function changes.
- **CellCNN vs. Quadratic:** adds a quadratic term and its L2 penalty.
- **Prototype vs. Quadratic:** different geometry, initialization and regularization; not an isolated alpha comparison.
- **Limitation:** variants were developed after earlier results. More splits do not create new independent donors.

Regularization: CellCNN on W, V ; Quadratic on W, Q, V . Prototype: output L2 plus $10^{-3} \text{mean}((a - 1)^2)$.

Quadratic Hard and Soft both reach 0.913 mean network AUC; learnable soft pooling doesn't increase it here.

All 100 planned splits are reported. Code, checkpoints, tables and figures are documented together.